
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is a core requirement for robust and reliable
2 machine learning systems deployed in open-world environments. Recent work
3 shows that unlabeled data can substantially strengthen OOD detection, but in-the-
4 wild data is intrinsically mixed: it contains both in-distribution (InD) and OOD
5 samples, and therefore does not provide a clean outlier set for detector training.
6 We introduce Medix, a median-centric framework that decisively converts this
7 mixed unlabeled signal into candidate outliers. Medix uses the median as a robust
8 central tendency operator whose resistance to noise and contamination makes it
9 a natural mechanism for separating atypical gradients from InD behavior. With
10 the extracted outliers and labeled InD data, Medix trains a strong binary OOD
11 classifier. Theoretically, we derive error bounds showing that Medix maintains a
12 low filtering error under the stated assumptions. Empirically, Medix consistently
13 outperforms existing methods in open-world settings, aligning the observed gains
14 with the theory and establishing median-based filtering as a powerful route for
15 unlabeled OOD detection.

16 1 Introduction

17 Real-world deployment immediately confronts machine learning models with safety- and reliability-
18 critical inputs that may fall outside the training distribution. Such out-of-distribution (OOD) samples
19 originate from unknown categories and should be rejected rather than forced into an in-distribution
20 label. Standard neural networks remain vulnerable in this setting because they typically lack an
21 explicit mechanism for recognizing and handling OOD inputs in practice (Nguyen et al., 2015).

22 OOD identification at inference time is therefore indispensable, but it is difficult precisely because
23 the unknown test distributions are absent during supervised training. Recent methods (Katz-Samuels
24 et al., 2022a; Du et al., 2024a) make an important step by leveraging additional “in-the-wild” data
25 to improve OOD detection. In particular, Katz-Samuels et al. (2022a) use unlabeled wild data to
26 regularize model training while preserving the primary goal of labeled InD classification. This
27 data source is compelling because it can be collected directly from the operating environment
28 after deployment, making it more representative of the OOD phenomena the detector will actually
29 encounter and thereby enabling more robust detection.

30 The central difficulty is that unlabeled wild data is not a clean OOD resource; it is a contaminated
31 mixture of InD and OOD samples. Because the detector observes only this mixed distribution and
32 receives no sample-level indication of whether an input is InD or OOD, a successful method must
33 first extract a reliable outlier signal from ambiguity. This problem remains sharply underdeveloped,
34 particularly from a theoretical perspective. To the best of our knowledge, Du et al. (2024a) is the only
35 prior work that provides a formal foundation for the “in-the-wild” setting, leaving a clear need for
36 principled alternatives.

Median-based methods have recently demonstrated strong robustness in data pruning (Acharya et al., 2024). These results point to a direct opportunity: the median can serve as a principled, contamination-resistant anchor for extracting OOD structure from unlabeled mixtures. This paper addresses the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question with a new median-centric perspective on OOD detection. We propose a median-based optimization framework and instantiate it as Medix, an algorithm that identifies OOD samples from wild data with low filtering error. Our analysis gives one of the few theoretical foundations for unlabeled “in-the-wild” OOD detection and shows why median-based filtering is robust in contaminated mixtures. Specifically, we bound the fraction of InD samples incorrectly flagged as outliers through two explicit terms: a contamination effect, which captures the influence of OOD points and remains controlled when the OOD proportion is below 50%; and a concentration effect, which uses the sub-Gaussian behavior of InD gradients to bound their deviations from the InD mean gradient. Together, these terms show that, with high probability, Medix keeps InD misclassification small, even under the worst-case contamination considered by the theory.

We evaluate Medix against two broad categories of competitors: methods trained only on InD data and methods trained with both InD data and auxiliary unlabeled data. On CIFAR-100 (Krizhevsky et al., 2009), Medix delivers a decisive FPR95 gain over the strong KNN+ baseline (Sun et al., 2022), improving the average by 40.98%. Unlike Outlier Exposure (Hendrycks et al., 2019), which assumes a clean auxiliary unlabeled dataset that is separable from InD data, Medix achieves stronger results without requiring such a restrictive distributional assumption. Relative to WOODS (Katz-Samuels et al., 2022a), Medix reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

Our key contributions are as follows:

- C1)** We introduce Medix, a median-centric greedy framework that filters outliers from unlabeled wild data and then trains an OOD detector on the extracted outliers together with InD samples. The central methodological contribution is the filtering stage, which converts mixed deployment data into usable OOD supervision.
- C2)** We prove theoretical guarantees for median-based filtering in unlabeled mixtures. The bounds show that both InD samples incorrectly flagged as outliers and OOD samples incorrectly retained as inliers are controlled by contamination and concentration effects: contamination remains bounded when the OOD proportion is below 50%, while concentration captures the stability of InD gradient behavior.
- C3)** We evaluate Medix across eleven InD-OOD pairs against 20 competitive baselines. Medix outperforms all reported baselines across these settings, and the outlier extraction experiment achieves a low error rate (e.g. 12.5%; see Figure 2), directly corroborating the theory.

2 Preliminaries and Problem Setup

This section fixes the OOD detection setting used throughout the paper and formalizes the data model, architectures, losses, and learning objective.

Labeled In-Distribution Data. Let $\mathcal{X}$ be the input space and $\mathcal{Y} = \{1, \dots, K\}$ the InD label space. The labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ contains n independently and identically distributed (i.i.d.) samples from $\mathbb{P}_{\mathcal{X}\mathcal{Y}}$, an unknown joint distribution on $\mathcal{X} \times \mathcal{Y}$. We denote its input marginal by $\mathbb{P}_{\text{in}}$, the InD distribution used to train the base model from $\mathcal{S}_{\text{in}}$.

Out-of-distribution detection. We study the practical deployment regime in which a model is trained on labeled InD data and later encounters inputs from classes absent from training, i.e., labels $y \notin \mathcal{Y}$. For these OOD inputs, the desired behavior is rejection rather than forced classification. Thus, inference reduces to a decisive binary question: whether an input belongs to the InD distribution or arises from an OOD source.

Unlabeled wild data. Labeled OOD samples are scarce because the potential OOD space is broad and expensive to annotate. We therefore incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, following the realistic deployment setting of Katz-Samuels et al. (2022a). This wild dataset is a

readily collected mixture of InD and OOD samples obtained after a model has been trained on $\mathcal{S}_{\text{in}}$. Following Du et al. (2024a); Katz-Samuels et al. (2022a), we characterize its marginal distribution with the Huber contamination model (Huber, 1964):

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π is the contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. The case $\pi = 0$ contains no OOD samples and is therefore trivial for the detection problem considered here.

Models and Loss Functions. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier with parameters $\phi \in \Phi$. Its output represents a soft probability distribution over the $K = |\mathcal{Y}|$ InD classes, and its supervised loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we use a separate binary classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameters $\theta \in \Theta$. Its binary loss is $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, with $y_+ > 0$ denoting InD and $y_- < 0$ denoting OOD.

Learning objective. The learning objective is to use both $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$ to train the OOD detector g_θ together with the multi-class classifier f_ϕ . At test time, we evaluate the resulting models with the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X} \times \mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ is the OOD test distribution.

3 Method: Median-Centric Framework for OOD Detection

This section presents **Medix**, a median-centric learning framework that uses unlabeled wild data as a concrete resource for OOD detection rather than treating it as incidental noise. Medix directly addresses the limitations of methods that rely only on InD data and is designed for open-world deployment, where models routinely face previously unseen inputs. The framework has two essential stages: **1) Outlier Extraction**: a filtering procedure that isolates candidate OOD samples from unlabeled wild data (Section 3.1), and **2) Detector Training**: training a binary OOD detector using labeled InD data and the extracted outlier candidates (Section 3.2). For stage 2, we follow the protocol introduced by Du et al. (2024a). Together, these stages turn an ambiguous unlabeled mixture into actionable OOD supervision and provide the basis for robust deployment in dynamic open-world settings.

3.1 Extracting Candidate Outliers from the Wild Data

Medix isolates candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$ through an optimization-based criterion built on model-parameter gradients. These gradients are computed from the classifier f_ϕ , trained only on the InD dataset $\mathcal{S}_{\text{in}}$, so deviations from the InD gradient profile become an explicit signal for OOD extraction. We now formalize this procedure.

Reference gradient estimation from InD data. The first step estimates a reliable InD reference gradient from $\mathcal{S}_{\text{in}}$. We train f_ϕ on $\mathcal{S}_{\text{in}}$ by empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, we compute the InD mean gradient $\bar{\nabla}_{\text{in}}$ by averaging loss gradients with respect to the model parameters over the InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

This vector $\bar{\nabla}_{\text{in}}$ is the reference gradient against which Medix measures how strongly other samples depart from InD behavior.

Motivation. The key empirical motivation is direct: as the number of OOD samples in $\mathcal{S}_{\text{wild}}$ increases, the median gradient of the wild set should move farther from the InD reference gradient $\bar{\nabla}_{\text{in}}$. We test

124 this behavior using CIFAR-10 (Krizhevsky et al., 2009) as InD data and SVHN (Netzer et al., 2011)
 125 as OOD data. The initial $\mathcal{S}_{\text{wild}}$ contains 10,000 CIFAR-10 samples disjoint from the training set used
 126 to learn $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN OOD samples and measure the
 127 L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of wild-set gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

128 Figure 1 shows a clear monotonic increase in this deviation, confirming that the median-gradient
 129 displacement tracks OOD contamination. This behavior is the central design signal behind Medix. It
 130 also yields the stopping rule: the filtering iteration terminates when the L_2 -norm deviation between
 131 consecutive iterations falls below the threshold ϵ , as detailed next.

132 **Filtering potential outliers from unlabeled wild data.** Motivated by Figure 1, we cast outlier
 133 extraction as the following optimization problem for identifying $\mathcal{S}_{\text{out}}^*$ within $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

134 Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted
 135 label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$.
 136 The optimization problem identifies a subset $\mathcal{S}$ in
 137 $\mathcal{S}_{\text{wild}}$ whose gradient EWM is closest to the InD
 138 average gradient $\bar{\nabla}_{\text{in}}$. Guided by Figure 1, the
 139 resulting subset $\mathcal{S}_{\text{in}}^*$ is treated as the InD-consistent
 140 portion of the wild data, while $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$
 141 provides the corresponding OOD candidate set.

142 Solving (4) exactly is computationally prohibitive
 143 as $\mathcal{S}_{\text{wild}}$ grows. Medix therefore uses a leave-one-
 144 out greedy approximation, summarized in Algo-
 145 rithm 1, that directly targets samples whose removal
 146 most strongly restores agreement with the InD gra-
 147 dient profile.

148 At each iteration, Medix computes the EWM of the
 149 remaining wild gradients and compares it with $\bar{\nabla}_{\text{in}}$.
 150 Let d_t denote the L_2 distance between $\bar{\nabla}_{\text{in}}$ and the EWM gradients of the remaining wild set $\mathcal{S}$, i.e.,
 151 $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. Medix then removes each candidate sample $i \in \mathcal{S}$ in a leave-one-out
 152 calculation, recomputes $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$, and measures the resulting distance to $\bar{\nabla}_{\text{in}}$. The score
 153 $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ quantifies how much sample removal improves InD alignment,
 154 so the k samples with the largest drops are selected as OOD candidates. The iteration stops when no
 155 removal produces a meaningful drop or when the maximum number of iterations is reached. The
 156 convergence rule uses the change in L_2 distance across iterations and halts when it falls below ϵ . This
 157 rule follows the monotonic trend in Figure 1: after enough OOD samples have been removed, the
 158 distance decreases to a small value, and continuing would risk removing InD samples and degrading
 159 performance.

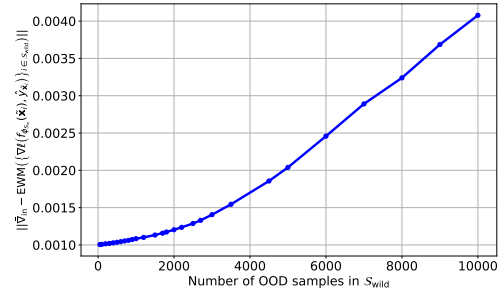


Figure 1: Median-gradient distance increases as OOD samples are added to $\mathcal{S}_{\text{wild}}$, motivating the Medix stopping criterion.

160 3.2 Training the OOD Detector with Candidate Outliers

161 After Medix identifies the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$, we train the OOD detector g_{θ} to sharply separate
 162 labeled InD data from the extracted outlier candidates, following Du et al. (2024a). The objective
 163 enforces a zero-threshold decision boundary: labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ receive positive outputs,
 164 while candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$ receive negative outputs. The loss is:

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \quad (5)$$

165 Because the 0/1 loss is non-differentiable, we optimize a differentiable sigmoid-based binary loss
 166 as a smooth surrogate that preserves the intended decision structure. We also include the InD risk

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

167 from Equation 2, which protects InD predictive accuracy while the detector learns the OOD boundary.
 168 This unified objective strengthens the generalization of g_θ and supports robust detection of OOD
 169 samples drawn from $\mathbb{P}_{\text{out}}$, as confirmed empirically.

170 4 Theoretical Analysis

171 We now give the theoretical guarantees for the Medix filtering stage. The theorems provide upper
 172 bounds on misclassification rates for both InD and OOD points, establishing the two-sided robustness
 173 of EWM filtering under the stated assumptions. Detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

174 Theorem 4.1 makes the source of the inlier error explicit. The bound separates a *concentration*
 175 *term*, which shrinks with sample size and captures finite-sample fluctuations in InD gradients, from
 176 a *contamination term*, which measures how OOD points can influence the EWM. Crucially, the
 177 guarantee remains controlled whenever the contamination ratio satisfies $\pi < 0.5$, formalizing the
 178 robustness of median-based filtering under substantial but non-majority OOD presence. We next
 179 address the complementary failure mode: *how many true outliers fail to be flagged and are mistakenly*
 180 *retained?* The following theorem bounds this OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

181 Theorem 4.2 completes the two-sided picture by bounding the fraction of OOD points mistakenly
 182 retained as InD. Taken together, Theorems 4.1 and 4.2 establish that Medix controls both filtering
 183 errors under the stated assumptions. The two rates are governed by three interpretable effects:

184 **Contamination Effect.** Theorem 4.1 shows that the median-based criterion remains stable even
 185 when the wild dataset contains a substantial OOD fraction, provided $\pi < 0.5$. The term $\pi/[2(1 - \pi)]$
 186 captures exactly how OOD gradients can skew the EWM as contamination increases. Conversely,
 187 Theorem 4.2 includes the penalty $\frac{1-\pi}{2\pi}$, which makes explicit the harder regime where OOD points
 188 are underrepresented and therefore harder to isolate.

189 **Concentration Effect.** Both bounds use the sub-Gaussian behavior of gradient coordinates to convert
 190 finite-sample variability into explicit error terms. For InD data, concentration around $\bar{\nabla}_{\text{in}}$ keeps the
 191 median anchored to the InD reference. For OOD data, concentration around μ_{out} quantifies the
 192 chance that an OOD gradient appears deceptively InD-like. These concentration effects decay as
 193 $1/\sqrt{m}$, where m is the relevant class sample size.

194 **Separation Effect.** This term, unique to Theorem 4.2, states the gradient separation needed for
 195 reliable OOD rejection. The exponential term $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ captures the trade-off directly:
 196 larger separation between OOD and InD mean gradients sharply reduces the probability that OOD
 197 gradients evade the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering achieves robust separation of InD and OOD samples in unlabeled data mixtures. The fraction of InD samples misclassified as outliers is bounded by contamination and concentration effects, while the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results provide rigorous theoretical assurance that Medix minimizes both types of errors under mild assumptions.

198 *Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.* The assumption that gradients
 199 of InD samples in each coordinate are sub-Gaussian is crucial for deriving concentration bounds
 200 and ensuring robustness in statistical estimation. Further empirical evidence as shown in Figure 4a
 201 confirms this assumption, where we observe that the histogram of gradient values for InD samples is
 202 bell-shaped and concentrated around the mean. This indicates that the gradients exhibit light tails,
 203 a defining characteristic of sub-Gaussian random variables, in which variables have exponentially
 204 decaying tails, meaning extreme gradient values are rare. This aligns with the observed behavior in
 205 the histogram. Furthermore, Figure 4b compares the empirical quantiles of the gradient distribution
 206 with the theoretical quantiles of a Gaussian distribution in a Q-Q plot (Wilk and Gnanadesikan, 1968).
 207 The close alignment of points with the 45-degree reference line demonstrates that the empirical
 208 distribution of gradients indeed closely resembles a Gaussian distribution. Since Gaussian random
 209 variables are a subset of sub-Gaussian variables, this supports the sub-Gaussian assumption. *We also*
 210 *provide a looser version of these bounds, which removes the sub-Gaussian assumption and holds*
 211 *under merely bounded second moments. This version is presented in Theorem C.3 (Appendix C.3).*
 212 *While the rates degrade, the core robustness guarantee of Medix still holds in this broader setting.*

214 5 Experiments

215 This section demonstrates the effectiveness of Medix across diverse InD–OOD dataset pairs and
 216 benchmarks it against 20 widely used baselines. The experiments are designed to test both standard
 217 OOD detection performance and the added value of extracting outliers from unlabeled wild data. All
 218 experiments are performed on hardware equipped with NVIDIA A100-SXM4-80GB GPUs. The
 219 necessary code to reproduce our results will be released upon acceptance.

220 5.1 Models, Datasets, and Baselines

221 **Datasets.** We use the same experimental protocol as Katz-Samuels et al. (2022a), which introduced
 222 the problem of learning OOD detectors with wild data, enabling a fair comparison to prior work.
 223 Specifically, we use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing, we select
 224 a suite of natural image datasets, including PLACES365 (Zhou et al., 2017), SVHN, TEXTURES
 225 (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) as OOD datasets ($\mathbb{P}_{\text{out}}$). To
 226 simulate wild data ($\mathbb{P}_{\text{wild}}$), we combine a subset of InD data ($\mathbb{P}_{\text{in}}$) with the OOD data ($\mathbb{P}_{\text{out}}$) under a

default mixing parameter $\pi = 0.5$. For example, when using PLACES365 as an OOD test set, we construct a wild mixture by combining CIFAR with PLACES365 as wild data and test on PLACES365 as the OOD set. This procedure is repeated across all OOD datasets and baselines. The InD CIFAR dataset is split into two halves: the first 25,000 images to train $\phi_{S_{in}}$, while the remaining images to generate the wild mixture S_{wild} . For gradient computation, we use the penultimate layer weights, as these have been shown to be particularly informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We evaluate using three standard metrics: (1) False Positive Rate (FPR95 $\downarrow$) of OOD samples when the True Positive Rate of InD samples is 95%, (2) Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy (InD Acc $\uparrow$).

Baselines. Our comparison encompasses a diverse set of competitive baselines, categorized based on whether they are trained using only InD data or both InD and wild data. For the methods trained exclusively on InD data ($\mathbb{P}_{in}$), we compare Medix against a variety of established OOD detection methods, including Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022); these methods use a model trained with softmax cross-entropy loss. Additionally, we also compare against methods based on contrastive loss, such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022), for a more comprehensive comparison. For methods that leverage both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize the training by promoting lower confidence or higher energy on outlier data. We also include a comparison with WOODS (Katz-Samuels et al., 2022a), which introduced the concept of wild unlabeled data and utilizes it for OOD detection through a constrained optimization approach. Finally, we included more recent baselines, including CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), to provide a more thorough evaluation.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), we use a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2 for the InD classifier $\phi_{S_{in}}$. We train it with stochastic gradient descent, momentum 0.9, weight decay 0.0005, initial learning rate 0.1, cosine learning rate decay, batch size 128, dropout rate 0.3, and 100 epochs. For Medix, we select ϵ from $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and k from $\{4k, 7k, 10k, 20k\}$, accounting for dataset size and optimizing OOD performance. For the OOD classifier g_θ , we initialize from the pre-trained InD classifier $\phi_{S_{in}}$ and add a linear binary-classification layer on the penultimate-layer features. We set the learning rate to 0.001 and fine-tune for 100 epochs according to Equation (5). The training loss combines binary classification with InD classification, with weight 10 on the binary component; all other training parameters match those used for $\phi_{S_{in}}$.

Table 1: Medix delivers the strongest OOD detection performance on CIFAR-10 as InD data. Results are averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using $\mathbb{P}_{in}$ only												
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
	Using $\mathbb{P}_{in}$ and $\mathbb{P}_{out}$												
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

5.3 Results

Table 2 gives the main CIFAR-100 results and shows that Medix outperforms every reported OOD detection baseline. Two findings are especially clear. First, methods that use both InD and wild data

Table 2: Medix delivers the strongest OOD detection performance on CIFAR-100 as InD data. Results are averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets										InD ACC $\uparrow$	
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average	
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29
Using $\mathbb{P}_{in}$ only												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10

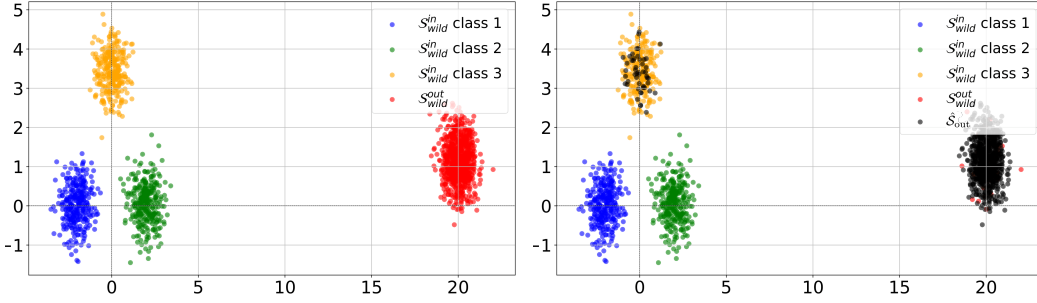


Figure 2: Medix cleanly extracts outliers from unlabeled wild data. (a) InD data S_{wild}^{in} and OOD data S_{wild}^{out} in the wild, with inliers sampled from three multivariate Gaussian distributions. (b) Outliers S_{out} filtered by Medix (in black), where the error rate from InD contamination S_{wild}^{in} is only 12.5%.

are substantially stronger than methods trained only on InD data, confirming the value of deployment-collected unlabeled data. Against KNN+, Medix reduces FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES. Second, Medix improves over competing wild-data methods: on CIFAR-100, it reaches an average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining an InD accuracy of 73.33%. This InD accuracy should be interpreted in light of the setup: Medix trains on 25,000 labeled InD samples, whereas baselines that do not leverage wild data use the full 50,000-image CIFAR-100 training set. Table 1 shows the same conclusion on CIFAR-10. Medix surpasses all baselines and reduces FPR95 relative to WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES, demonstrating that the median-based filtering advantage transfers across InD datasets.

A representative visual example of Medix. We further examine Algorithm 1 (Outlier Extraction) on a controlled 2-dimensional synthetic example that makes the filtering behavior visually transparent. The InD data consists of three Gaussian classes (with means $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$), each using covariance $0.25 \cdot I$. The OOD data follows $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that the outliers selected by Medix align closely with the ground truth OOD region. Medix flags 87.5% of actual OOD samples as outliers, providing a direct visual demonstration of robust outlier extraction from S_{wild} .

Additional studies and insights. Appendix A provides additional evidence for the same claim: Medix is both effective and practically stable. The appendix includes hyperparameter sensitivity analysis (Appendix A.2), where Medix remains robust across tested values of ϵ and k ; an EWM-versus-geometric-median comparison (Appendix A.1), where EWM provides the more sensitive filtering signal; a broader comparison with competitive contrastive and recent baselines (Appendix A.3), where Medix maintains a substantial advantage; a large-scale unseen OOD evaluation with $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4); a compute and memory profile (Appendix A.6); a pseudo-label quality study showing resilience to noisy or low-confidence labels (Appendix A.5); and a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has become a central research direction (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). A major line detects OOD data through scoring functions that assess distributional fit, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Sehwal et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A second major line improves OOD behavior through training-time regularization (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). These methods reduce model confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data, but most require an auxiliary OOD dataset.

A smaller but highly relevant line incorporates unlabeled wild data (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) (Katz-Samuels et al., 2022a; Geng et al., 2025), but much of this work does not provide a direct outlier extraction mechanism. Du et al. (2024a,b) introduce explicit filtering, yet their thresholding strategy is fundamentally different from our median-centric extraction rule. Moreover, Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, where each batch has a prescribed InD/OOD ratio; large outsourced datasets typically do not provide this structure because samples are mixed randomly at the dataset level. Medix directly handles dataset-level mixing without relying on batch composition. Positive-unlabeled learning methods (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022) are also related, but they primarily distinguish $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ rather than simultaneously extracting outliers and training an OOD detector. Medix fills this gap with a median-centric outlier identification method with provably low error rates.

7 Conclusions

This work establishes Medix as a median-centric framework for OOD detection with unlabeled in-the-wild data. By exploiting the intrinsic robustness of the median, Medix filters outliers from mixed unlabeled data and turns them into effective supervision for a reliable OOD detector. The theoretical analysis proves bounds on inlier misclassification, showing that Medix remains robust under substantial OOD contamination (up to 50%) through explicit sub-Gaussian concentration and contamination terms. We also provide complementary limits on OOD misclassification under clear separation conditions. Across diverse benchmarks, Medix decisively outperforms 20 baselines: it reduces average FPR95 by 40.98% relative to strong baselines such as KNN+ and improves over state-of-the-art methods including WOODS and DRL, while reaching an outlier extraction error as low as 12.5%.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

We next justify a key design choice in Medix: using the element-wise median rather than the geometric median (Acharya et al., 2024) for outlier detection. We answer this through a preliminary experiment with CIFAR-10 as InD data and SVHN as OOD data. The wild set $\mathcal{S}_{\text{wild}}$ contains 5k CIFAR-10 samples disjoint from the training set used to train $\phi_{\mathcal{S}_{\text{in}}}$, which provides $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples and track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the wild-set median gradient and the proportion of OOD samples removed. Figure 3 shows that, for the same number of OOD samples, the element-wise median identifies a substantially higher proportion of OOD samples as outliers than the geometric median. This result supports the element-wise median as the more sensitive and reliable filtering operator for Medix.

Table 3: Medix remains stable across the tested hyperparameters ϵ and k for OOD detection.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We evaluate the sensitivity of Medix to ϵ and k using CIFAR-100 as InD data and SVHN as OOD data. Table 3 shows that Medix is robust across the tested choices of both hyperparameters. With $\epsilon = 0.005$ and $k = 20000$, Medix achieves an FPR95 of 0.16. When ϵ is reduced to 0.0005 or k is halved to 10000, performance remains strong, with FPR95 values of 0.20 and 0.68, respectively. The degradation is graceful rather than abrupt, showing that Medix does not depend on a fragile hyperparameter setting to outperform the baselines. This stability strengthens Medix as a practical deployment method, especially when exhaustive tuning is infeasible.

A.3 Comprehensive comparison with recent competitive methods

We extend the comparison to strong baselines with diverse and contrasting learning objectives, including SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). These methods are not based on the Medix objective, making them a useful stress test for the proposed framework. We also include recent methods CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c) for a broader competitive evaluation. For this experiment, we use CIFAR-100 as the InD dataset and train a ResNet-34 model¹ following Ming et al. (2023); Zhang et al. (2024). The model is trained with stochastic gradient descent, momentum 0.9, weight decay 10^{-4} , 500 epochs, initial learning rate 0.5 with cosine scheduling, and batch size

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

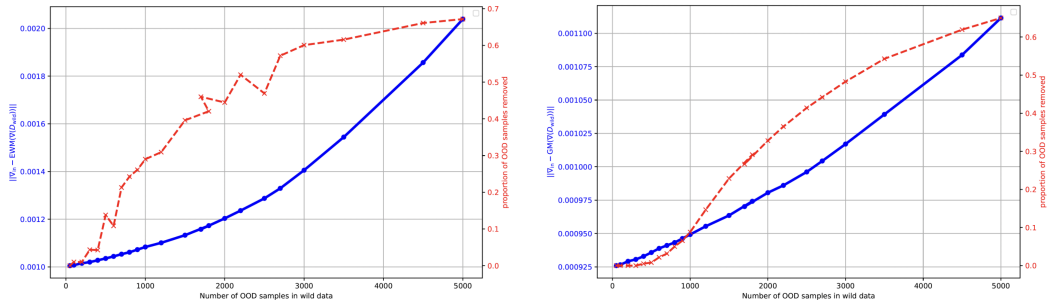


Figure 3: Element-wise median (EWM) provides a stronger filtering signal than geometric median (GM) in the Medix diagnostic study.

Table 4: Medix remains competitive against recent strong baselines on CIFAR-100 as InD data. Results are averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

512. We use the checkpoints provided by Ming et al. (2023)². Table 4 shows that Medix outperforms
528 these competitive baselines by a substantial margin, including methods trained with contrastive
529 objectives, reaching an average FPR95 of 8.68% and an average AUROC of 97.76%. Medix also
530 provides a provably low filtering error, giving it theoretical support that most alternatives lack.

631 A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

632 We next test Medix in a large-scale unseen OOD setting that directly stresses generaliza-
633 tion beyond the wild OOD source. The setting is demanding for two reasons: it uses
634 a large wild OOD dataset ($\mathbb{P}_{\text{out}}$), increasing data complexity and computation, and it eval-
635 uates on a test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) that deliberately differs from the wild OOD
636 data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$. This protocol is a stronger proxy for deploy-
637 ment, where future OOD samples need not match the unlabeled wild data collected earlier.
638 We use CIFAR-100 as labeled InD data, 300K
639 Random Images Hendrycks et al. (2019) as un-
640 labeled wild OOD data, and SVHN as the test
641 OOD dataset. Because 300K Random Images
642 differs substantially from SVHN, this experi-
643 ment evaluates whether Medix learns a trans-
644 ferable OOD signal rather than merely mem-
645 orizing the wild source. Table 5 shows that
646 Medix achieves the best result among wild-data
647 baselines, with FPR95 41.29 ± 1.2 and AUROC
648 87.25 ± 0.6 , confirming that the filtering strategy
649 remains effective under this harder unseen-OOD shift.

Table 5: Medix improves OOD detection on large-scale unseen OOD data.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

650 A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

651 We also test whether pseudo-label quality, measured through softmax confidence for $\hat{y}_{\tilde{x}_i}$, materially
652 affects Medix filtering and whether a simple pre-filtering step can improve robustness. We use
653 CIFAR-10 as InD data and LSUN-Resize as OOD data. We compute softmax probabilities for
654 each pseudo-labeled sample and remove low-confidence predictions using a threshold of 0.6, which
655 discards 15.98% of the training samples.

656 Performance is essentially unchanged: both variants achieve FPR95 of 0.01%, and AUROC changes
657 only from 99.98% to 99.99% with filtering. This result shows that Medix is resilient to noisy or low-
658 confidence pseudo-labels, and that additional confidence filtering is unlikely to provide meaningful
659 gains in this setting.

²<https://github.com>

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We further verify that the Medix filtering phase is computationally feasible and memory-efficient on large datasets. We profile Medix on an NVIDIA A100-SXM4-80GB GPU with approximately 15,000 unlabeled samples for each InD–OOD pair. The experiment uses CIFAR-10 and CIFAR-100 as InD datasets and LSUN-Resize as the OOD dataset. Table 6 shows modest GPU memory use and tractable wall-clock time, even with thousands of unlabeled samples. These measurements confirm that Medix filtering is computationally practical and does not impose excessive memory overhead at this scale.

Table 6: Medix filtering remains computationally tractable on an NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix differs sharply from semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024). For example, OpenML Saito et al. (2021) focuses on handling outliers during semi-supervised learning (SSL) for a standard classifier, whereas Medix explicitly detects OOD samples from unlabeled wild data and trains a dedicated OOD detector. Thus, SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) improve classification despite outliers, while Medix targets open-world OOD detection when labeled OOD data is unavailable.

The objective is also different. SSL methods typically treat OOD samples as noise to suppress during training, while Medix actively identifies and uses these samples to build a stronger OOD detection system.

The technical mechanism is distinct as well. OpenMatch relies on consistency regularization, enforcing stable classifier outputs under stochastic transformations and encouraging smooth decision boundaries. Medix instead uses median-based filtering, exploiting the median’s robustness to extreme values to extract OOD candidates. The accompanying theory bounds the error through contamination and concentration effects, giving Medix a direct guarantee for the filtering task.

B Broader Impacts and Limitations

Reliable OOD detection is essential for deploying machine learning systems in high-impact real-world environments. Medix contributes a median-centric framework that strengthens OOD handling and directly supports safer model deployment. Its practical relevance extends to domains such as healthcare, autonomous systems, and finance, where rejecting unfamiliar inputs is often as important as classifying familiar ones. By improving OOD detection from unlabeled deployment data, Medix addresses a core barrier to trustworthy machine learning. Future work can build on this foundation by integrating Medix with generative models for outlier synthesis or adapting it to dynamic environments where OOD distributions evolve over time.

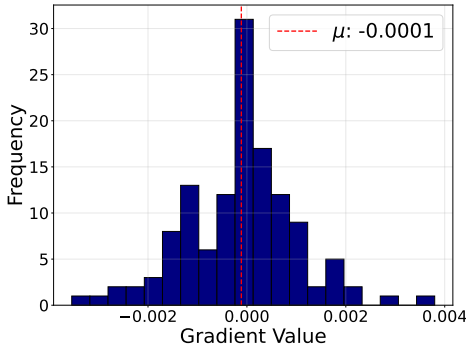
The main limitation of Medix is its reliance on an unlabeled mixture of InD and OOD data, which may be noisy, corrupted, or inconsistently sampled in practice. This assumption is not a weakness unique to Medix; it captures the uncertainty inherent in deployment-time data and is a standard starting point in robust learning theory. We make the assumption explicit and testable through the main theorems and the geometric separation condition in Remark 4.3. We further strengthen the framework in Theorem C.3 by removing the need for sub-Gaussian tails, showing that the guarantee persists under weaker distributional assumptions. Thus, Medix relies on assumptions that are interpretable, relaxable, and useful as a principled foundation for OOD detection under realistic noise.

701 C Proof of Theorem

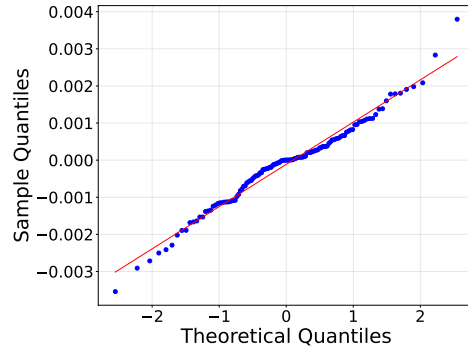
702 Table 7 consolidates the notation used in the theoretical and empirical analysis. It covers gradient
 703 quantities, sample sizes, filtering terms, and the parameters entering the concentration bounds. Unless
 704 stated otherwise, boldface denotes vectors, $\mathcal{X}$ is the input space, and all gradients are computed with
 705 respect to a fixed pretrained model f_ϕ .

Table 7: Notation used in the Medix analysis. Bold symbols denote vectors and $\mathcal{X}$ denotes the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients exhibiting sub-Gaussian behavior in each coordinate. (left) Histogram of gradient values (CIFAR-100 InD data) showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, confirming alignment with sub-Gaussianity.

706 **C.1 Bound on the Inlier Misclassification Rate**

707 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 708 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 709 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 710 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

711 *Proof.* We structure our argument into two key steps. In **Step 1**, we establish a high-probability
 712 concentration bound for the gradients of InD samples around their mean, denoted as $\bar{\nabla}_{\text{in}}$. We define
 713 an InD sample whose gradient lies within this high-probability region as a *good inlier*. In **Step**
 714 **2**, we leverage a “swapping” argument on the EWM objective to demonstrate that removing more
 715 than a $(2\eta + \frac{\pi}{2}(1-\pi))$ fraction of *good inliers*—where η represents the fraction of InD samples
 716 with gradients outside the high-probability region (i.e., non-good InD samples)—would contradict
 717 optimality. Combining these two steps, we derive the stated upper bound on the inlier misclassification
 718 rate, ERR_{in} .

719 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 720 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 721 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

722 Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

723 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

724 so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

725 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

726 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 727 $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

728 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

729 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 730 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

731 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

732 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 733 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

734 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

735 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

736 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
737 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

738 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

739 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

740 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

741 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

742 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
743 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
744 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
745 called *good inliers*.

746 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 747 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

748 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 749 thereby contradicting the optimality of $\mathcal{S}^*$.

750 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 751 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

752 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

753 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

754 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

755 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 756 available for swapping in.

757 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 758 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

759 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

760 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

761 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

762 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

763 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 764 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

765 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

766 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

767 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

768 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

769 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

770 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
771 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

772 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
773 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
774 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

775 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

776 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
777 increase it, so the median distance from μ_j does not increase and possibly decreases.

778 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

779 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

780 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
781 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

782 □

783 C.2 Bound on the Outlier Retention Rate

784 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
785 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
786 (Appendix C.1), but instead of relying on inlier concentration, it exploits a separation assumption
787 between the OOD and InD gradient means.

788 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 789 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

790 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

791 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

792 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

793 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

794 *Proof. Step 1: Concentration of OOD Gradients.*

795 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 796 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

797 has probability at most $p_{\text{out}}(\varepsilon)$.

798 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

799 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

800 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

801 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

802 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

803 which holds with probability at least $1 - \delta$.

804 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

805 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

826 **Differences with the original bound.** The sub-Gaussian result is stronger because it features a
 827 decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

828 that decreases exponentially with dimension d , making the bound extremely tight for large d . In
 829 contrast, the new bound, relying solely on the assumption of finite fourth moments, exhibits a decay
 830 term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

831 that decays as $O(1/d)$, so the guarantee is looser but applies to heavier-tailed gradients. This extension
 832 is important because it preserves a meaningful Medix guarantee under the weaker assumption of
 833 finite fourth moments. Both bounds retain the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

834 that shrinks as m_{in} increases, independent of the tail assumptions.

835 *Proof.* The proof follows the structure of the original Theorem 4.1, consisting of two main steps: (1)
 836 establishing a concentration bound on the fraction of “bad” InD samples (those far from ∇_{in}) without
 837 relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier misclassification
 838 rate. The key difference is the replacement of the sub-Gaussian tail bound with a moment-based
 839 bound derived from the finite fourth moment assumption.

840 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

841 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 842 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

843 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 844 S_{wild} .

845 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 846 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

847 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

848 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

849 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

850 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

851 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

852 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

853 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

854 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
855 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

856 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
857 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

858 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

859 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

860 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

861 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 862 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

863 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 864 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

865 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 866 thereby contradicting the optimality of $\mathcal{S}^*$.

867 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 868 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

869 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

870 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

871 Define the swap size as

$$m'_\epsilon := \min \{ m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

872 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 873 available for swapping in.

874 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 875 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

876 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

877 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

878 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

879 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

880 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 881 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

882 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

883 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

884 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

885 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

886 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

887 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 888 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

889 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 890 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 891 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

892 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

893 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 894 increase it, so the median distance from μ_j does not increase and possibly decreases.

895 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

896 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

897 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 898 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

899 □